

The master theorem

CS3230 · Lecture 4

Demo course material, written for OpenPapr

Recurrences of one shape

- $T(n) = a T(n/b) + f(n)$
- a subproblems, each of size n/b
- $f(n)$: the work to divide and combine

The critical exponent

- Compare $f(n)$ with $n^{\log_b a}$
- $n^{\log_b a}$ counts the leaves of the tree

Case 1

- $f(n) = O(n^{(\log_b a - e)})$ for some $e > 0$

$$T(n) = \Theta(n^{\log_b a})$$

- Leaves dominate

Case 2

- $f(n) = \Theta(n^{\log_b a})$

$$T(n) = \Theta(n^{\log_b a} \log n)$$

- Every level costs the same

Case 3

- $f(n) = \Omega(n^{\log_b a + \epsilon})$, plus regularity

$$T(n) = \Theta(f(n))$$

- The root dominates

Master theorem

- For $T(n) = aT(n/b) + f(n)$, compare $f(n)$ with $n^{\log_b a}$
- Three cases: leaves, every level, or root dominates

Examples

- $T(n) = 2T(n/2) + n$: Case 2, $\Theta(n \log n)$
- $T(n) = 8T(n/2) + n^2$: Case 1, $\Theta(n^3)$
- $T(n) = 2T(n/2) + n^2$: Case 3, $\Theta(n^2)$

When it does not apply

- $T(n) = 2T(n/2) + n \log n$ falls in a gap
- Use the recursion tree or substitution

Summary

- Find a, b, f
- Compare $f(n)$ with $n^{\log_b a}$
- Pick the case, check regularity for Case 3